



Project Proposal: NakPom (អ្នកភូមិ)

1. Executive Summary

NakPom is a comprehensive community ecosystem designed to strengthen the digital fabric of Cambodian society. By integrating family connectivity with civic engagement, the app serves as a central hub for communication, problem-solving, and information sharing.

- **The Problem:** Modern fragmentation has weakened traditional community ties. Families lack private digital spaces, students struggle to give honest feedback to institutions, and the government often lacks real-time, localized data on citizen needs. So, how does the app address that problem?
- **The Solution:** NakPom provides a multi-layered platform where users can securely connect with their "Krousa" (family), participate in anonymous community forums, and access educational resources, while providing a direct data-driven link to organizations and government bodies.

2. Target Audience

- **Primary Users:**
 - **All Khmer citizens**, from youth to the elderly, with a focus on high **Khmer Language** accessibility and intuitive UI.
 - **Education Sector:** Students, teachers, and school administrators for feedback loops and announcements.
 - **Corporate/NGOs:** Organizations looking to boost internal productivity and staff engagement.
 - **Government:** Local and national authorities seeking to conduct surveys and understand citizen sentiment for urban and rural development.
- **Key Needs:** Deep Khmer language support, high privacy (anonymity features), and an easy-to-use interface for non-technical users.

3. Core Features

- **Neati Krousa (Family & Community Circles):** Dual-layered networking. A private "Internal" space for immediate family and a "Community" space for broader village or neighborhood interaction.
- **Neati Sarpheap (The Anonymous Confession Wall):** A safe space for members to report issues, confess problems, or provide feedback to schools and employers without fear of social repercussions.
- **Neati Chamness (Knowledge Hub):** A curated library of "Soft Skills," short courses, and educational videos aimed at personal and professional growth.

- **Neati Chumroeurn (Citizen Feedback):** A dedicated module for government surveys and community polls to streamline national development planning.

4. Technical Stack

The tools and technologies that will power the application.

- **Frontend: Jetpack Compose** (Native Android) for a modern, fluid, and responsive user experience.
- **Backend: Kotlin** (Ktor or Spring Boot) for a high-performance, type-safe server environment.
- **Database: MySQL** for robust relational data management and secure user records.
- **Environment:** Hosted on **Linux servers** to ensure maximum stability, security, and cost-efficiency.

5. Project Roadmap

A high-level timeline of development.

- **Phase 1: Design & Prototyping:** UI/UX wireframing focused on Khmer typography and designing the relational database schema.
- **Phase 2: Core Development:** Building the Kotlin API and implementing the Neati Krousa and Neati Sarpheap modules in Jetpack Compose.
- **Phase 3: Testing & Debugging:** Rigorous security audits and "Beta Testing" with a small group of Khmer students and families.
- **Phase 4: Deployment:** Official launch on the Google Play Store and initial onboarding of government/school partners.

6. Potential Challenges

Acknowledge risks and how you plan to mitigate them.

- **Scalability & Cost:** To manage a growing user base with minimal cost, we will utilize Docker containers on Linux to optimize server resources and implement caching strategies to reduce database load.
- **Security:** To protect sensitive "Sarpheap" (anonymous) data, we will implement End-to-End Encryption for private messages and strict JWT (JSON Web Token) authentication. Regular penetration testing will be conducted to safeguard against external attacks.